Applied Corporate Finance

Week #1 – Review of valuation techniques

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Plan of Attack

- Overview of Corporate Finance Functions

- Discounted Cash Flow Valuation Methods:
  - Free Cash Flow:
    - WACC
    - Valuation Using Adjusted Present Value
    - Comparing APV and WACC
  - Capital Cash Flow.
  - Equity Cash Flow.

- Valuation Using Multiples.

- Valuation and uncertainty

- Example

What is Corporate Finance?

- Aspects that have financial implications for a corporation
- Defined broadly everything a corporation does fits under corporate finance
- For example:
  - Whether to invest in a new production line or close an existing one?
  - Whether to engage in a marketing campaign?
  - Whether to invest R&D to build a new product?
  - Whether to hire more workers?
  - Whether to invest in human capital of current workers?
  - Whether to enter a new market?
  - Etc…

Why is Finance Director such a central role?

- Analyzes how current performance compares to budget.
- Recommends current adjustments for the departments of the company.
- Forecasts the future of the company.
- Evaluates investments return.
- Analyzes the best combination of funding for the investments.
- Supports the investment decisions.
- **All this and much more to maximize the firm value.**

Decisions in Corporate Finance

- **Investment Decision**
  - Valuation
  - Considering Risks and Returns

- **Financing Decision**
  - Find the right financing structure for the project or firm
  - In-house vs. separate entity
  - How much Debt
  - What kind of Debt

- **Payout Decision**
  - How and when to payout to investors

The Goals of Corporate Finance

- The broad goal of corporate finance is to **maximize firm value**
- A more narrow goal is to **maximize shareholders value**
- This view may not be as narrow as it sounds. In an attempt to maximize profits firms will have to care for other stakeholders:
  - ESG goals
  - Consumers may punish firms that have unethical business practices
  - Firms may have to treat their employees well due to competition

A Famous Theorem

- Franco Modigliani and Merton Miller in a series of articles (1958, 1961, 1963) show that **under certain conditions finance is irrelevant**.
  - Both received Nobel prizes for their contributions.

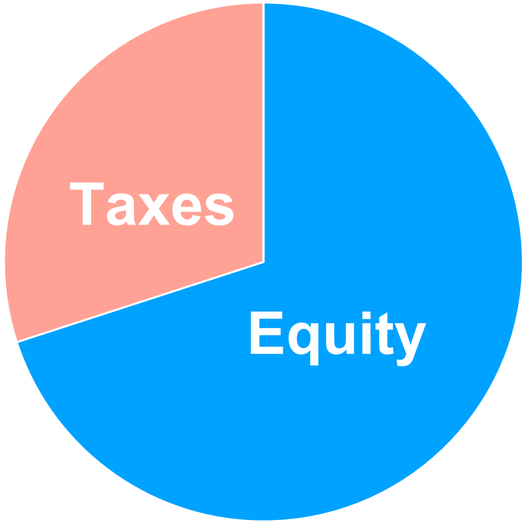
- **The MM Theorem assumes:**
  - No Transaction Costs.
  - No Taxes.
  - No Costs of Financial Distress.
  - No Agency Costs.
  - No Asymmetric Information.

- Use it as a road map to understand **how finance can create value**.

Example: Tax Deductibility of Interests

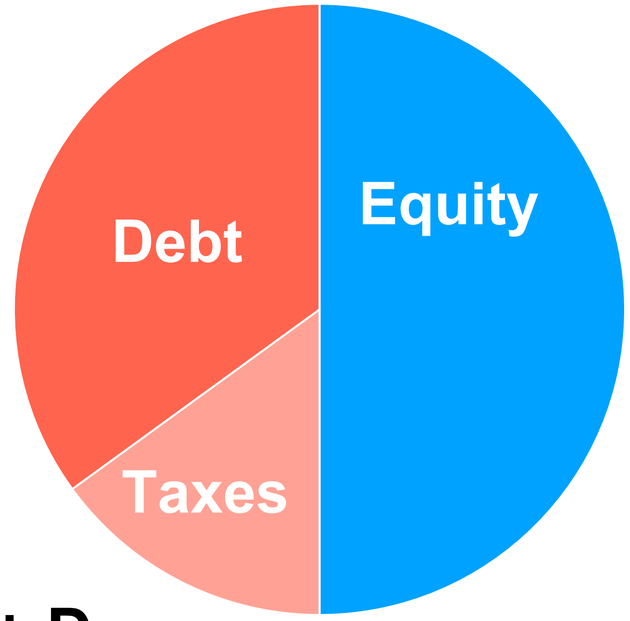
	Unlevered Firm	Levered Firm
1. Earnings before interest and taxes	1 000 000	1 000 000
2. Interests	0	400 000
3. Earnings before taxes	1 000 000	600 000
4. Taxes (35%)	350 000	210 000
5. Profits	650 000	390 000
Cash Flow to Shareholders and Debtholders (2.+5.)	650 000	790 000

Capital structure – with taxes



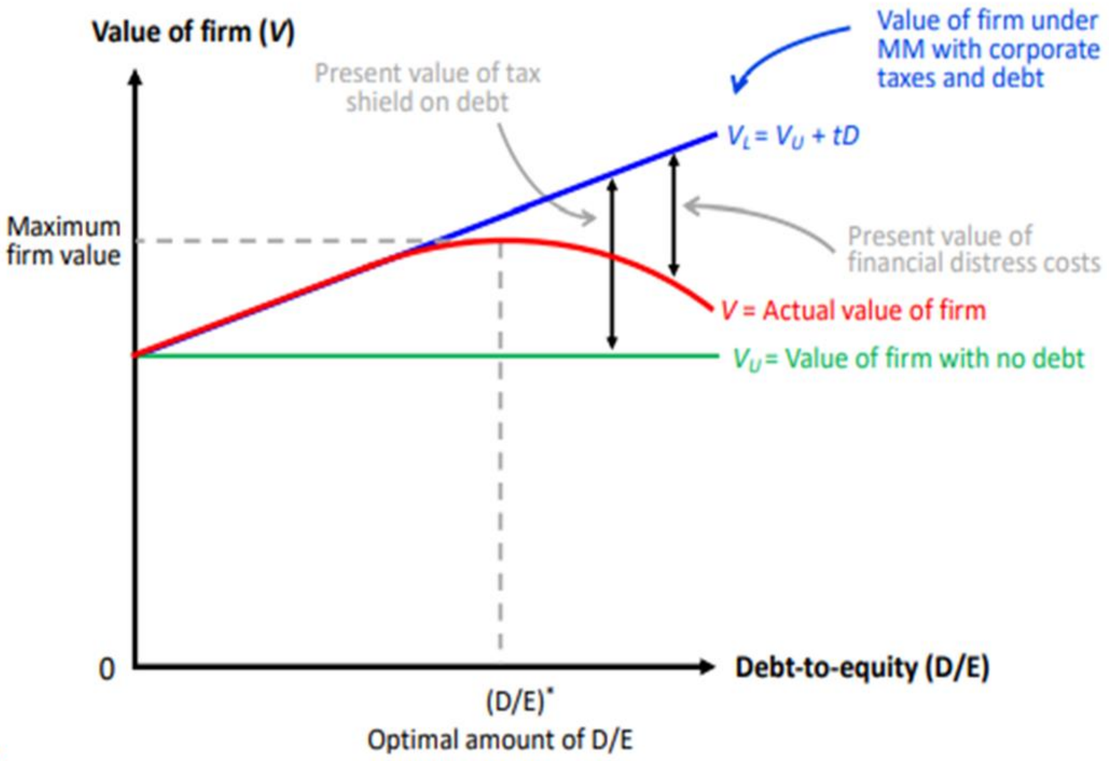
$V_U = E$

$V_U < V_L$



$V_L = E + D$

Capital structure



- There is a trade-off between tax advantage of debt and costs of financial distress:

$$V_L = V_U + \text{PV (Tax Shield)} - \text{PV (Financial Distress Costs)}$$

- In theory, there is an optimal capital structure: Leverage should be taken until the tax benefits are more than offset by the financial distress costs

Objective for the next few hours

- We know how to value a **100% equity financed project**.

- Now we want to take financing into account. Recall **financing matters** through:
  - Tax shields.
  - Costs of financial distress.
  - Agency Costs.

- There are **different techniques in the valuation world**. Much more than the ones we will cover in this course.

- We want to review the basics of:
  - Discounted Cash Flows (DCF): WACC, APV, Capital Cash Flows (CCF), Equity Cash Flows (ECF); and
  - Multiples/Comparables.

Financial value of an asset

"The financial value of any asset or project corresponds to the **present value of its expected future cash flows.**"

Two Different FCF (Free Cash Flows) Methods

- Two different methods for total firm valuation with financing (with free cash flows):
  - Weighted Average Cost of Capital (WACC).
  - Adjusted Present Value (APV).

Valuation Using WACC (Weighted-Average Cost of Capital)

WACC Method

1. Determine the free cash flow (FCF) of the investment.
2. Compute the weighted average cost of capital (WACC).

$$r_{WACC} = \frac{E}{E+D} r_e + \frac{D}{E+D} r_d (1-\tau)$$

3. Compute the value of the investment/company by discounting the FCF of the investment/company by the WACC

$$V_{L,0} = \frac{FCF_1}{1+r_{WACC}} + \frac{FCF_2}{(1+r_{WACC})^2} + \cdots + \frac{FCF_T}{(1+r_{WACC})^T}$$

4. Terminal Value: $$TV_T = \frac{FCF_T \text{ x } (1+g)}{r_{WACC} - g}$$

WACC Method (cont.)

- FCF = Cash flow from operations + Cash flow from investments

- FCF = EBIT x (1-τ) [or NOPAT] + D&A – ΔNWC – Capex + other

Revenue
– COGS (Cost of Goods Sold)
= Gross Profit
– Operating Expenses (e.g., SG&A)
= EBITDA
– D&A (often included in SG&A)
= EBIT/Operating Profit/Operating Income
– Taxes
= NOPAT
+ D&A
– Capx
– ΔNWC
= Free Cash Flow (FCF)

Note:

- Non-cash expenses should be summed-up to the FCF.
  - For example, impairments and provisions should be added back since they are a non-cash expense.
- Non-cash income should be subtracted from the FCF.
  - For example, accounting gains on the sale of fixed assets.

WACC Method (cont.)

- $\frac{D}{E+D}$ **or** $\frac{D}{E}$ **- leverage ratio**
  - You should use market values of debt and equity, sometimes book values are used.
  - If the company changes the leverage ratio the WACC changes
  - Some companies have a target or relative stable leverage ratio
    - More usual in mature companies.
    - This assumption can be strong in some companies

- τ **- corporate tax rate**
  - You should use marginal tax rate
    - Not easy to calculate when the companys pays taxes in different countries
  - Sometimes we use effective tax rates
    - Use an historical average
    - Be careful about carry forward losses carry-forward
    - See if there is volatility in historical effective tax rates

WACC Method (cont.)

- ***D* and *E* are market values.**
  - If no market value for *D*, use book value.
  - Note: using book values for the debt of highly levered or distressed firms may cause problems.

- **In *D*, only include interest bearing debt.**
  - Include interest bearing short-term and long-term debt.

- **Correct the effect of cash:** If the company holds a large amount of cash, you need to remove the **excess cash** from the calculation to get a beta for just the business.

Which Discount Rate in APV? (cont.)

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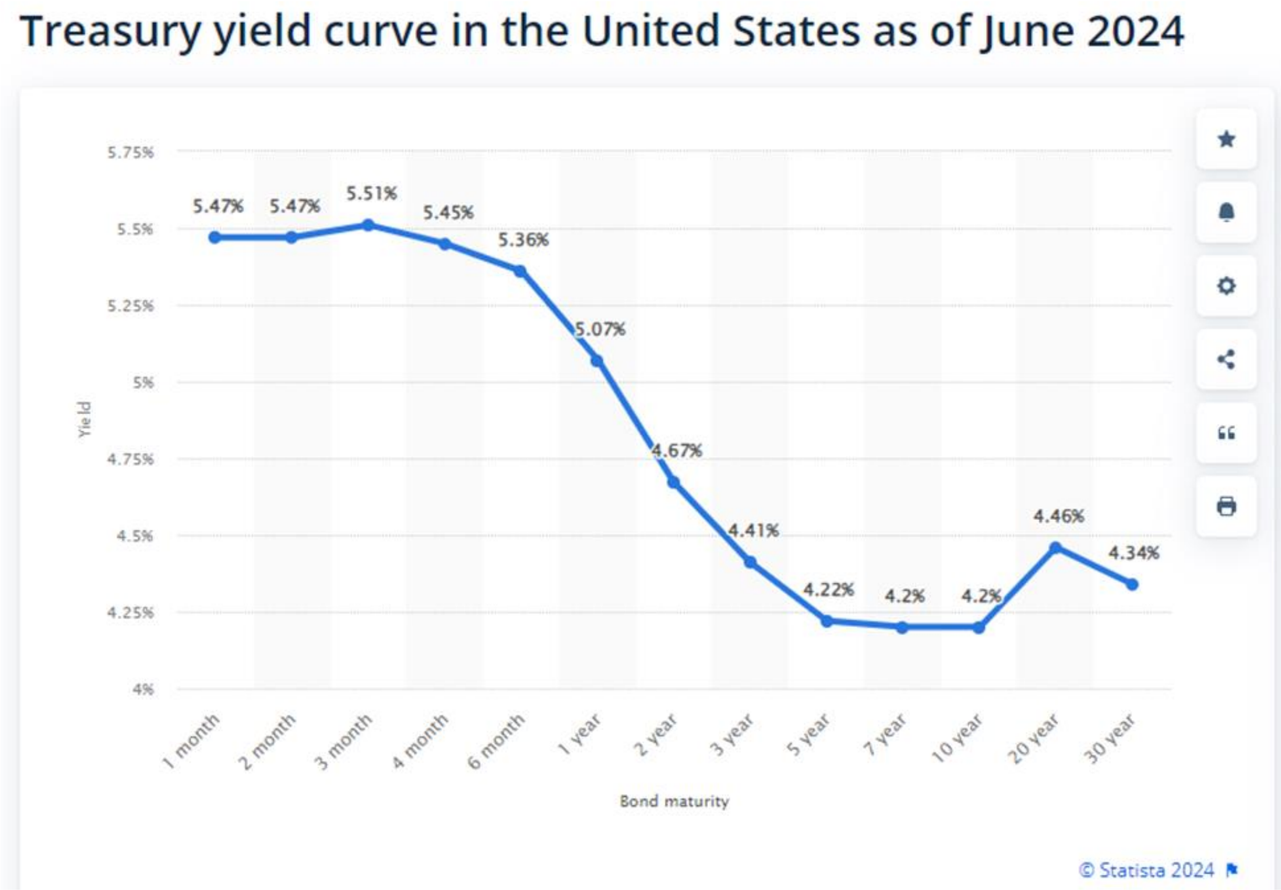
WACC Method (cont.)

- **Cost of debt** $(\boldsymbol{r_d})$
  - Return on a traded, long-term obligation of the company
  - Typically use:
    - Quoted spread over treasuries for given debt ratings.
    - YTM of comparable bond (though YTM > true r_d)
    - Computed β_d
- **Cost of equity** $(\boldsymbol{r_e})$
  - Typically use $r_e = r_f + \beta_e(MRP)$:
    - firm's CAPM β_e (if capital structure has been stable; target a D/V ratio)
    - Industry β_A, re-levered to match capital structure
- **Beta of assets** $(\boldsymbol{\beta_A})$
  - If the firms targets a D/V ratio (it is reasonable to assume that $\beta_{TS} = \beta_A$), then: $\beta_A = \frac{D}{V}\beta_d + \frac{E}{V}\beta_e$

Risk-free rate

- How much is the risk-free rate?
  - Short-term or long-term for r_f?
  - Historical average or forward looking r_f?



Equity Risk Premium

- Historical Equity Risk Premium
  - There are plenty of estimates of risk premium of market over Treasury Bonds. Historically between 5% to 6%.
  - Ibbotson & Associates (2000): Realized average risk premium over treasuries in 1999 was 9.32%.

- Implied Equity Risk Premium
  - Fama & French (2002): Risk premium implied by fundamentals and stock prices over the 1951-2000 time period was between 2.55% and 4.32%.
  - Kaplan and Ruback (1996): Risk premium implied by sample of MBOs was 7.8%

- Equity Risk Premium – Survey Estimates
  - Graham and Harvey (2001) survey of CFOs: 10-year risk premium ranges between 3.6% and 4.7%.
  - Welch (2001) survey of finance professors: 1-year risk premium averaged 3.4%, 30-year premium averaged 5.5%.

Country	Equity Risk Premium
Angola	13,00%
Argentina	20,34%
Australia	4,33%
Austria	4,86%
Brazil	7,67%
Bulgaria	6,46%
Czech Republic	5,13%
Denmark	4,33%
Finland	4,86%
France	5,13%
Germany	4,33%
Greece	7,67%
India	7,26%
Ireland	5,13%
Italy	7,26%
Japan	5,27%
Mexico	6,87%
Netherlands	4,33%
New Zealand	4,33%
Norway	4,33%
Poland	5,46%
Portugal	5,93%
Spain	6,46%
Sweden	4,33%
Switzerland	4,33%
United Kingdom	5,13%
United States	4,33%

Source: NY U Stern, Damodaran (Jan 2025)

Discount Rate

- Best check on the discount rate:
  - **Sensitivity Analysis.**
  - This applies to any of the estimated inputs of a model.

- **Getting "fancy" with the discount rate is a low payoff activity.**
  - Using "term structure" (different risk-free rates for each period) adds very little.
  - Using Fama-French (three-factor, four-factor or five-factor) model instead of CAPM may not be easy.

- **Calibration can be helpful.**
    - How do we proceed with calibration?

Discount Rate (cont.)

Three Basic Steps on Calibration

- Value Company as it is.
  - Use current forecasts.
  - Find the discount rates.

- Is estimated value close to the market value?
  - No? Try to understand why.
  - Make adjustments to get the market value.

- Apply discount rate/growth rate to:
  - Other comparable companies/investments.
  - Check whether assumptions seem sensible in other valuations.

Terminal Values

There are generally three ways to proceed to compute TV

- If project is liquidated or sold at the end of the horizon, we calculate its ***salvage value*** → **"Liquidation method"**
  - Make sure you account for any taxes on capital gains or TS on capital losses.

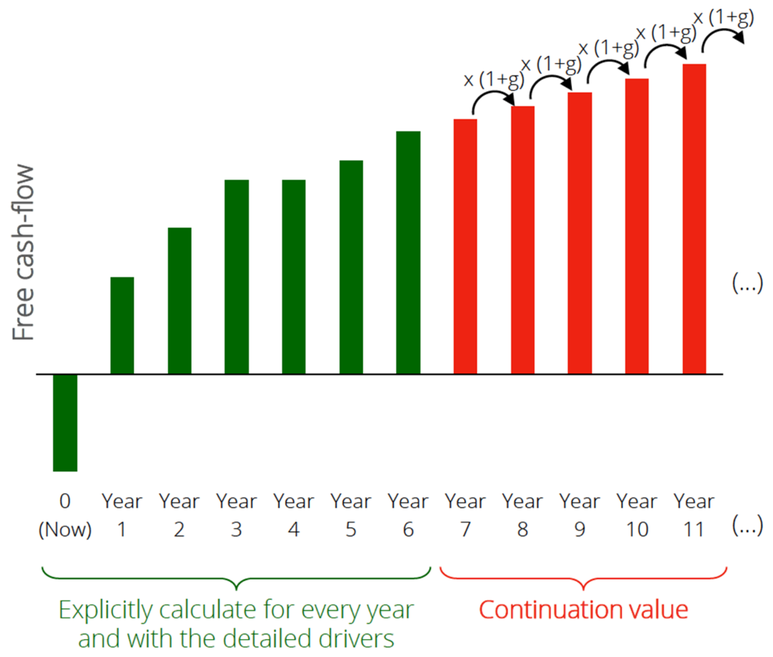
- Take cash flow (that presumably you already calculated earlier) and calculate the ***continuation value*** → **"Perpetuity method"**

- Assume that you can sell the firm using a simple rule involving **earnings multiples** → **"Exit Multiples method"**

Terminal Values – Continuation Value

Continuation Value

Incremental cash-flows generated by an investment



Perpetual cash-flows after explicitly calculated period

$$CV = \frac{FCF_{T+1}}{r-g}$$

Stream of limited cash-flows after explicitly calculated period

$$CV = \frac{FCF_{T+1}}{r-g}\left[1-\left(\frac{1+g}{1+r}\right)^{N}\right]$$

Terminal Values: Perpetuity Calculation

- For the Perpetuity Method, we assume the firm is in steady state.
  - May not be appropriate with high-growing companies, cyclical industries, concessions.

- Use the last free cash flow, grow it by one period (recall PV formula), and divide it by: $r_{WACC} - g$.

$$TV_T = \frac{FCF_T \text{ x } (1 + g)}{r_{WACC} - g}$$

- Where do we get "g"?

- Ask yourself: are depreciation, NWC and other flows you have in the cash flow estimate consistent with the "g" that you choose?

Terminal Values: Multiples of Earnings

- If you use a multiple of earnings to calculate the sale price, then you need to calculate earnings – not free cash flows.

- Where do you get the multiple from?
  - Pick the multiple using comparable firms.
  - Be sure to perform sensitivity analysis.
  - More on Valuation with Multiples later.

Valuation Using Adjusted Present Value

The Adjusted Present Value Method

- An alternative approach to the WACC is to compute the Adjusted Present Value (APV).

- The three simple steps involved in computing the APV are:

  - **Step 1**: Value of the project as if all-equity financed: use the after-tax cash flows and discount them at the cost of capital.
    - Remember that for an all-equity firm the cost of capital equals the cost of equity.

  - **Step 2**: Add the present value of the tax shield (TS) generated by the project and other effects of debt.

  - **Step 3**: Account for other financing effects (if any), such as **costs of financial distress, agency costs or loan subsidies**.

$$\mathbf{APV = NPV_{all\ equity} + PV(Financing\ Choices)}$$

The Adjusted Present Value Method

- To be clear:

$$\mathbf{APV = NPV_{all\ equity} + PV(TS) - PV(CFD) +/- PV(other\ effects)}$$

- APV is also known as a **valuation by components.**

- In the simpler form, we consider only the Tax Shield:

$$\mathbf{APV = NPV_{all\ equity} + PV(TS)}$$

The Adjusted Present Value Method (Cont.)

- This is simply MM Proposition I with taxes in action.

$$\mathbf{APV = V_L = V_U + PV[TS]}$$

- The APV method uses the value additivity principle to evaluate the contribution of both cash flows and increased debt tax shields.
  - It can easily be adapted to include other financing side effects.

- We do this to separate the **value of running the business** from the **value created by financing**.

- Doing this allows us to identify the **sources of value and to discount different risks appropriately**.

Which Discount Rate in APV?

- You need the rate (r_A or r_U) that would be appropriate to discount the firm's cash flows if the firm were 100% equity financed.

- This rate is the **expected return on equity if the firm were 100% equity financed**.

- To get it, you need to:
  - **If** the project is in line with the firm's current operations **and** there are data available (listed company with stable leverage ratio), can use own beta, and unlever it.
  - Find comparables, *i.e.*, publicly traded firms in same business.
    - Estimate their expected return on equity if they were 100% equity financed, by unlevering the equity betas.

Discount Rate in APV? (cont.)

Industry Name	Beta	D/E Ratio	Effective Tax rate	Unlevered beta	Cash/Firm value	Unlevered beta corrected for cash
Advertising	0,97	43,69%	11,90%	0,73	10,13%	0,82
Aerospace/Defense	1,19	14,17%	12,32%	1,07	6,76%	1,15
Air Transport	1,45	84,52%	18,11%	0,88	10,15%	0,98
Auto & Truck	1,63	145,39%	10,43%	0,78	15,62%	0,92
Beverage (Alcoholic)	0,61	58,59%	13,68%	0,42	2,75%	0,44
Beverage (Soft)	0,65	33,20%	16,70%	0,52	4,72%	0,55
Broadcasting	1,03	41,82%	14,09%	0,78	12,02%	0,89
Building Materials	1,02	19,89%	16,26%	0,89	4,33%	0,93
Cable TV	1,14	308,53%	8,22%	0,34	13,96%	0,40
Chemical (Basic)	0,66	59,31%	9,66%	0,46	7,52%	0,49
Coal & Related Energy	1,29	10,72%	9,42%	1,19	16,33%	1,42
Computer Services	1,01	12,53%	15,35%	0,92	5,65%	0,98
Drugs (Biotechnology)	1,33	14,73%	1,72%	1,20	7,00%	1,29
Drugs (Pharmaceutical)	1,14	17,14%	7,65%	1,01	3,29%	1,04
Education	0,67	27,01%	13,88%	0,56	3,97%	0,58
Electrical Equipment	1,29	13,33%	8,59%	1,17	4,78%	1,23
Electronics (Consumer & Office)	1,19	9,80%	5,72%	1,11	5,18%	1,17
Electronics (General)	1,24	15,85%	10,24%	1,11	4,69%	1,16
Entertainment	0,77	20,18%	6,40%	0,67	8,42%	0,73
Environmental & Waste Services	1,19	70,25%	13,93%	0,78	11,63%	0,88
Farming/Agriculture	0,62	104,72%	13,22%	0,35	5,84%	0,37
Furn/Home Furnishings	0,82	92,21%	13,57%	0,48	13,63%	0,56
Green & Renewable Energy	0,81	96,87%	11,93%	0,47	5,43%	0,50
Healthcare Products	1,31	21,92%	6,46%	1,12	2,53%	1,15
Healthcare Support Services	0,80	64,72%	16,24%	0,54	5,28%	0,57
Hospitals/Healthcare Facilities	0,61	240,03%	15,38%	0,22	5,58%	0,23
Hotel/Gaming	1,28	33,23%	12,81%	1,02	6,75%	1,09
Metals & Mining	1,08	36,46%	7,01%	0,85	8,14%	0,92
Oil/Gas (Integrated)	0,78	56,69%	33,65%	0,55	15,76%	0,65
Oil/Gas (Production and Exploration)	0,96	80,97%	10,90%	0,59	13,97%	0,69
Oil/Gas Distribution	0,56	76,22%	7,42%	0,36	8,35%	0,39
Paper/Forest Products	0,69	37,98%	13,35%	0,54	7,43%	0,58
Power	0,70	84,03%	17,22%	0,43	8,49%	0,47
Precious Metals	1,14	24,51%	10,94%	0,96	6,50%	1,03
Real Estate (Development)	0,71	63,16%	8,06%	0,48	11,57%	0,54
Real Estate (General/Diversified)	0,98	166,26%	7,54%	0,43	3,25%	0,45
Real Estate (Operations & Services)	0,93	130,88%	8,97%	0,47	4,64%	0,49
Retail (General)	0,96	39,01%	12,46%	0,74	8,39%	0,81
Rubber& Tires	1,01	46,97%	7,87%	0,75	10,09%	0,83
Semiconductor	1,62	19,55%	7,23%	1,41	6,70%	1,51
Shipbuilding & Marine	0,83	48,56%	6,72%	0,61	12,49%	0,70
Software (Entertainment)	1,18	7,16%	8,97%	1,12	7,43%	1,21
Software (Internet)	1,17	45,32%	9,79%	0,87	3,24%	0,90
Software (System & Application)	1,06	6,00%	8,05%	1,01	4,60%	1,06
Steel	1,05	62,15%	9,07%	0,71	23,36%	0,93
Telecom (Wireless)	0,77	168,04%	17,57%	0,34	8,72%	0,37
Telecom. Equipment	0,90	18,82%	7,62%	0,79	19,75%	0,98
Telecom. Services	0,91	106,39%	14,95%	0,51	5,26%	0,53
Tobacco	0,58	58,85%	12,06%	0,40	5,43%	0,42
Transportation	0,83	78,89%	15,64%	0,52	9,13%	0,57
Transportation (Railroads)	0,95	53,70%	15,04%	0,68	13,05%	0,78
Trucking	1,01	67,62%	20,55%	0,67	5,31%	0,71
Utility (General)	0,93	122,15%	20,32%	0,49	11,31%	0,55
Utility (Water)	0,87	149,86%	19,97%	0,41	6,54%	0,44

Source: Source: NY U Stern, Damodaran (Jan 2025)

Which Discount Rate and TV for APV (as all equity)?

- Use the comparable β_A as an estimate the project's β_A.

- Use the estimated β_A to calculate the all-equity cost of capital r_A

$$r_A = r_f + \beta_A * \text{Market Risk Premium}$$

- Use r_A to discount the project's FCF.

- Use the last free cash flow, grow it by one period (recall PV formula), and divide it by r_A - g.

$$TV_T = \frac{FCF_T \text{ x } (1+g)}{r_A - g}$$

Tax Shield of Debt

- Need to account for the debt tax shields, including any associated with the terminal value and discount these at the appropriate interest rate.

- Interest payments are tax deductible. Reduce tax payments by:

$$\tau_c * interest$$

- These deductions depend on the interest payments and the tax rate.
  - If the interest payments will appear on the financial statements, use them.
  - If only the debt levels appear, you need to translate them into implied levels of interest payments.

Important Caveats of Tax Shields

- **We only consider corporate taxation**
  - We could also consider our **investors' tax rates**, but we rarely do so

- **Debt Tax Shield is only valuable when we pay taxes**
  - Otherwise, must keep track of **NOLs and loss carrybacks**, but this becomes messy

Discounting the Tax Shield

- The choice of the discount rate depends on the risk. **What is the risk of the tax shield?**

- In general the tax shield will have its own risk, which will depend also on the probability that the government will change its tax policy, and similar issues.

- Precisely estimating the risk of the tax shield could be very difficult (and not worthwhile), thus typically we choose among two assumptions: **the risk of the tax shield is equal to the *risk of the debt* or the *risk of the assets*.**
  - Since these are assumptions there is no absolute right or wrong answer. However, one may claim that one criteria will probably be closer to the truth, **depending on circumstances**.

Discounting the Tax Shield (cont.)

- If debt is **predetermined or has a low level**, risk of tax shields equal to risk of repayments to debtholders; so r_D is the correct discount rate.

- For example, when there is a **permanent debt D**, we write the PV(TS) **present value of tax shields as** $t_C D$, which is obtained by discounting with r_D.

- If

  a. high level of debt (e.g. LBOs), or

  b. level of debt varies with firm value (**target a D/V ratio**),

  then riskiness of tax shields is similar to that of operating assets and tax shields should be discounted at r_A.

Two Possible Scenarios: r_A or r_d?

SCENARIO 1 – Using r_A

- Debt in the future is not fixed.
  - Debt and interest expense are tied to FCF.
  - The company has a **target D/V ratio.**

- Works well when big changes in capital structure,
  - LBOs (more highly leveraged transactions) + Bankruptcy.

- In those cases, the ability to use tax shields has more systematic risk than the ability to pay debt.

$$TS = \text{Tax Rate} * \left[\frac{Int_1}{(1+r_A)} + \frac{Int_2}{(1+r_A)^2} + \ldots + \frac{Int_T}{(1+r_A)^T} \right]$$

- Int_t comes from debt repayment schedule and interest rates.

- Beta of assets **(β_A)**: $\beta_A = \frac{D}{V}\beta_d + \frac{E}{V}\beta_e$

Two Possible Scenarios: r_A or r_d?

SCENARIO 2 – Using r_d

- You expect a **predictable, stable, and low level of debt**.

- Ability to use tax shield has the same systematic risk as the ability to pay debt.
  - Best suited in **lower-leverage situations.**

$$TS = \text{Tax Rate} * \left[\frac{Int_1}{(1+r_D)} + \frac{Int_2}{(1+r_D)^2} + \ldots + \frac{Int_T}{(1+r_D)^T} \right]$$

- For permanent debt (fixed amount) with the same risk as interest:

  TS = Tax Rate * Permanent Debt = τ * D

- Beta of assets **(β_A)**, considering $\beta_D = 0$: $\beta_A = \dfrac{\beta_e}{(1 + (1-\tau) \text{ x } \frac{D}{E})}$

Tax Shield and the Betas

- Whatever assumption you make about the risk of the tax shield, you have to take care to be **consistent** throughout the valuation.
  - In particular, remember that **the assumption you make about the risk of the tax shield also affects the equation you will use to lever and unlever the beta.**

- If the risk of the tax shield is equal to the risk of the *debt* (**D is constant and with low level**), then such equation is:

$$\beta_A = \beta_D \frac{D(1-t_C)}{E+D(1-t_C)} + \beta_E \frac{E}{E+D(1-t_C)}$$

- If instead the risk of the tax shield is equal to the risk of the *assets* (**D/V ratio is constant or high levels of debt**), then such equation is:

$$\beta_A = \beta_D \frac{D}{E+D} + \beta_E \frac{E}{E+D}$$

Summary of Tax Shield Discount Rates

- **Remark 1:** If debt is predetermined, **D is constant and has a low level**, risk of tax shield is similar to risk of payments to debt holders; so r_d is the correct discount rate.

- **Remark 2:** If high level of debt or if level of debt varies with firm value (**D/V is constant**), then the risk of the tax shield is similar to that of operating assets and tax shields should be discounted at r_A.

- **Remark 3:** Some firms have **high leverage or maintain a target leverage ratio while maintaining an investment grade rating on their debt.** For these firms, the correct discount rate is probably closer to r_d.

Summary of Tax Shield Discount Rates

- You have to add any debt tax shields on NPV regarding the terminal value.

- Let's see **2 different ways…**
  1. **If D stabilizes at a permanent** (and low) level, D_T.

$$\text{PV}(TS) = \frac{\tau * r_D * D_T}{r_D} = \tau * D_T$$

  2. **If (D/V) stabilizes at some level**, λ ($D_T = \lambda * V_T$):

$$TV(TS) = \frac{\tau * r_D * D_T}{r_A - g}$$

Extending the APV Method

- Other potential financing effects are loan guarantees, subsidized loans, costs of issuing securities, etc.

- You could also add the PV of the costs of financial distress, agency costs, etc. How?
  - Write scenarios and include costs of distress in bad scenarios.
  - Often people omit PV(costs of distress) and PV(agency costs) because they are very difficult to quantify.

- If you want to include any of these costs, simply calculate the present value and add or subtract it.

Distress and agency costs

Direct costs	**Legal expenses** **Court costs** **Advisory fees** **(...) Other**	• When facing financial distress and close to or in bankruptcy the company will have extra expenses that it would not have otherwise if it did not have debt. Increasing debt will increase these costs and/or the probability of occurrence
Indirect costs	**Reputation**	• Financial distress may affect demand, since the firm cannot commit to remain in business. Examples: (A) Products which may require spare parts and maintenance (e.g. automobiles), (B) Firms which depend on customer relationships (e.g. airlines) • Suppliers may stop sending raw materials • Firms lose flexibility when so closely monitored by creditors • Employees may be less willing to make firm-specific investments when the firm is in or near distress
	Fire sales	• Assets may be liquidated at fire sale prices because (A) Creditors have little incentive to search for prices higher than the value of their claims or (B) Firms with highly specific assets that cannot be sold quickly without significant losses in value will suffer most (e.g. oil drilling)
Agency costs	**Risk shifting**	• Shareholders may take high risk, negative NPV projects in the hope of realizing the upside potential, leaving bondholders to bear the downside risk (overinvestment)
	Debt overhang	• Shareholders may be unwilling to finance positive NPV projects when the firm is in financial distress since debtholders have priority over the CFs (underinvestment)
	Cashing out	• Shareholders may try to get money out of the firm ahead of higher priority claims (e.g. by paying cash dividends)

APV vs WACC

Comparing Methods: APV

- **Pros:**
  - Implicit assumptions are very clear. No contamination.
  - Works even if debt is not permanent.
  - Is very easy if the **level of future debt is known**.
  - **Clearer:** Puts the spotlight on what is creating value. Easier to track down where value comes from.
  - Assists in the **decision of how to structure financing** for projects.
  - **More flexible:**
    - Just add other effects as separate terms.
    - Allows to use different discount rates for different components

- **Cons:**
  - Requires explicitly **deriving schedule of future debt levels** and interest payments
  - It is increasingly used, but **still less than WACC**.

Comparing Methods: WACC

Pros:

- **Most widely used.**
- The inputs are usually relatively easy to get.
- If you have a precise **target debt-to-value policy**, then it is **easy and relative accurate**.
  - **this can be be highly restrictive.**
- Requires fewer, often **simpler computations**.

Cons:

- **Mixes up the effects of assets and liabilities.**
  - Errors/approximations in effect of liabilities contaminate the whole valuation.
  - **Does not reveal where the value is coming from.**
- **Not very flexible**. Cannot easily allow for changes in:
  - Other effects of financing (e.g., costs of distress, issue costs)
  - Non-constant debt ratios
  - Changes in tax rates

Practical Implications

- In principle, we can always **use either WACC or APV**. As long as you follow all the steps and you are careful in making the same assumptions **you should obtain (approximately) the same solution**.

- In practice, however, one of the two will be much simpler to use depending on the situation.
  - For example: **complex, changing or highly leveraged capital structure (i.e. LBOs), APV is much better.**

- **Let's consider two cases:**
  - **Debt is rebalanced.**
  - **Debt is predetermined.**

Practical Implications (cont.)

- If **debt is rebalanced** (*i.e.* the firm has a **target debt/asset ratio**):
  - **Computing WACC is much easier.**
  - APV becomes more complex since you do not know *D*.
  - **Bottom Line: often prefer WACC.**

- If **debt is predetermined or constant** (*i.e.* firm knows the evolution of *D*):
  - **APV is easy to compute** by discounting future expected interest payments.
  - WACC instead has a problem, because if the debt is not rebalanced then D/V changes over time and so does WACC.
    - you can always forecast D/V values, re-compute a different WACC for each year and discount. Lots of opportunities for errors.
  - **Bottom Line: prefer APV.**

Other DCF Valuation Methods

Three Cash Flow Valuation Methods

- The names for the three methods correspond to the type of cash flow that is used in the valuation:
  - **Free Cash Flow (FCF)** – Provides Enterprise Value.
  - **Capital Cash Flow (CCF)** – Provides Enterprise Value.
  - **Equity Cash Flow (ECF)** – Provides Equity Value.
- The three methods differ in their measure of cash flows and the discount rate applied to those cash flows.
- The three methods provide consistent valuations when applied correctly.

Free Cash Flow and Capital Cash Flow

- **FCF and CCF both**
  - **value the entire firm** and not just the debt or the equity.
  - are measures of **cash flow to all suppliers of capital**
    - Equity holders *and* debt holders.

- The *difference* between CCF and FCF is the treatment of the tax benefits of deductible interest payments (interest tax shield).

  Capital Cash Flow = FCF + Interest Tax Shield

- CCF uses all cash flows that are paid/payable to investors:

  CCF = Net after-tax income (NI) + Interest Expense (I) + Depreciation – Capex – Change in NWC

- When they are applied correctly, the two methods provide equivalent valuations.

Free Cash Flow and Capital Cash Flow (cont.)

- CCF method includes the benefits of the tax shields as cash flows.
  - The more the tax advantages, the higher the capital cash flow.
  - The discount rate for this method often uses the **return on assets**.

- FCF method includes the tax benefits either by
  - **Reducing the discount rate (WACC)**
    - The more the tax advantages, the lower the discount rate.
  - Separately **adding the PV of the tax shield (APV)**
    - In fact, when we use r_A as the discount rate of both terms:

$$APV = \frac{FCF}{r_A} + \frac{t_C r_D D}{r_A} = \frac{FCF}{r_A} + \frac{ITS}{r_A} = \frac{CCF}{r_A}$$

    - Thus, **APV = NPV(CCF) discounted at r_A**
      - Sometimes called **"Compressed APV".**

When Should We Use CCF?

- CCF valuation is generally **useful for similar situations as APV**:
  - Highly Leveraged Transactions (e.g. LBOs)
  - Firms in Financial Distress.
  - Bankruptcy.

Equity Cash Flow (ECF)

- ECF measure the cash flow available to stockholders after payment to debt holders are deducted from operating cash flows.

- Payment to debt holders are sometimes called “Debt Cash Flows”(DebtCF), and they include interest and principal payments.

ECF = CCF + DebtCF

- **As DebtCF are paid out of operating cash flows before equity cash flows, debt cash flows are safer than equity cash flows.**

- ECF are riskier than cash flow measures that combine DCF and ECF, which implies **higher discount rates.**

Equity Cash Flow (cont.)

- ECF are calculated by subtracting interest and adding debt variations to CCF.

ECF = CCF – Interest + Change in Debt

ECF = Net after-tax income (NI) + Depreciation – Capex – Change in NWC + Change in Debt

- The **discount rate used in the ECF is the *return on equity*** and the value we obtain is the ***equity value* of the firm**.

- Requires that a **new equity cost of capital** be calculated each year of the forecast **when leverage is changing**.
  - Lots of unlevering and levering beta

Overall Comparing Methods

- In general, we typically **prefer both APV and WACC to ECF**.

- APV is preferred to the ECF for many of the same reasons that APV is superior to WACC.
  - The use of one discount rate for **ECF requires** that the firm have a **constant debt to total capital ratio**.
    - If the debt to total capital ratio varies over time, we need to calculate a different discount rate for the equity for each year.
  - It is **much easier to make mistakes using the ECF**.
    - In particular, it is critical to include increases in debt in the cash flows, particularly in the terminal value calculation.

Valuation Using Multiples

Valuation Using Multiples

- The main idea is to value the target asset (firm) by using information on the pricing of other similar assets.
  - Also called “Relative” or “Comparables” Valuation

$$[\text{Value Measure/Key Characteristic}]_{\text{Ind.Average}} * \text{Key Characteristic}_{\text{Firm}}$$

- Numerator of multiple is typically the value of the firm
  - e.g., price, enterprise value.

- Denominator of multiple is the characteristic that is important for that firm, often industry-specific:
  - paid miles flown for airlines
  - revenues for pharmacies
  - number of patients for dialysis industry
  - clicks or subscribers for web site
  - number of patents for a hi-tech firm.

Valuation by Multiples: Implicit Assumptions

1. Comparable companies assumed to have expected **cash flows growing at the same rate** and have the **same level of risk** as the company being valued.

2. The value of the company is assumed to vary in direct proportion with changes in the performance measure; i.e. if expected **EBITDA increases by 8%, expected firm value also increases by 8%.**

⇒ If these assumptions are valid, valuing by multiples will likely be *more* accurate than the DCF approach because it incorporates current market expectations of cash flows and discount rates.

Motivation for Multiples

- Firms in the **same business should have similar multiples.**

- If the firm's actual FCF is a perpetuity:

MV firm = FCF/(WACC-g) => MV firm/FCF = 1/(WACC-g)

- Comparables will have a similar MV firm/FCF provided they:
  - Have the same **WACC (requires similar D/(D+E)).**
  - And are **growing at a similar rate**.

Types of Multiples

- Cash-flow-based *Value* multiples:
  - MV of firm (EV)/Earnings, MV of firm (EV)/EBITDA, MV of firm (EV)/FCF.

- Cash-flow-based *Price* multiples:
  - Price/Earnings (P/E), Price/EBITDA, Price/FCF.

- Asset-based *Value* multiples:
  - MV of firm (EV)/BV of assets, MV of equity/BV of equity.

- Industry-specific *Value* multiples:
  - MV of firm (EV)/revenues, MV of firm (EV)/Number of Customers, MV of firm (EV)/Hospital Beds.

Procedure

Implicit Assumption: Firms in the same business should have similar multiples.

- Step 1: **Identify firms in same business** as the firm you wish to value. Be careful not to induce a selection bias.

- Step 2: Calculate **the multiple(s) for comparable firms**.

- Step 3: Calculate **average for the set of comparable firms**.
  - E.g., value-weighting, equal-weighting, median, etc.
  - This gives an estimate of the multiple we can use in valuing our firm.

- Step 4: **Apply the multiple on the firm you wish to value**.

- Step 5: Often different multiples give you different answers:
  - **Make judgment reflecting what is economically reasonable**.

Remarks

- When choosing comparable firms, you face a bias-variance trade-off:
  - **too many firms** leads to firms that are **not truly comparable**;
  - **too few firms** means that your average will reflect **idiosyncrasies of those firms**.

- For firms with no earnings or limited asset base (e.g. hi-tech),
  - Price-to-patents multiples.
  - Price-to-subscribers multiples.
  - Or even price-to-Ph.D. multiples!

Remarks (cont.)

- For transactions, can also use **multiples for comparable transactions**.
  - Similar transaction values.
  - Transactions imply control of the company or not?
  - But be aware, everyone might be over/underpaying.

- Multiples based on equity value (or stock price, e.g., P/E) as opposed to total firm value ignore effect of leverage on the cost of equity (or assume the firms have similar leverage)
  - **Beware if comparables have very different leverage**.

Pros and Cons of Multiples

Pros:

- Incorporates simply a lot of information from other valuations in a **simple way**.
- Embodies market **consensus about discount rate and growth rate**.
- **Free-ride on market's information**.
- Can provide discipline in valuation process by ensuring that **your valuation is in line with other valuations**.
- Sometimes, **what you care about is what the market will pay**, not the fundamental value (tough it is always important to have an idea about the fundamental value!).

Cons:

- **Difficult to find *true* comparables.**
  - Implicitly assumes all comparables are alike in growth rates, cost of capital, and business composition.
- **Hard to incorporate firm-specific information.**
  - problematic if operating changes are going to be implemented.
- **Relies on accounting measures** being comparable.
- **Differences in accounting practices** can affect earnings and equity-based multiples.
  - Therefore better to use FCF and EBITDA multiples.
- **Book values can vary across firms** depending on age of PPE (Property, plant and equipment).
- **Market liquidity** can affect the multiples.
- **If market is overpaying, you will too!**

Bottom Line on Multiples

- **Multiples complement DCF methodology:**
  - Check on valuation; market-based perspective.
  - **Extremely useful when you do not have cash flow projections.**

- **EBITDA or cash-flow multiples are preferable to (net) earnings multiples.**
  - **More consistent treatment of leverage.**
    - Companies with different leverage will have different P/E multiples, even tough same business risk and growth.
  - **Less ability to manipulate**
    - Easier to manipulate net earnings through accountings than EBITDA or cash flow.

Bottom Line on Multiples (cont.)

- Because of the many limitations, **never rely on just a single multiple or on valuation based only on multiples.**

- Best to use multiples only as a **check for the valuation based on discounted cash-flows**.
  - After you have done a throughout valuation, you can compare your predicted multiples.
  - If your predicted multiples are out of line then you have to convince yourselves (and your clients) that your model is reasonable.

Bottom Line on Multiples (cont.)

- Because of accounting differences, be **careful in using multiples to compare firms across industries**, and especially, **across countries.**

- As a rule of thumb when choosing the basis for multiples, remember that:
  - The higher up the basis is in the income statement (e.g. **sales**) the **less it is subject to changes in the accounting method.**
  - On the other hand, the **less it reflects differences in operating efficiency across firms** (e.g. firm's pricing policies, production efficiency, etc.).

Valuation

After Valuation is Done…

- Three possible answers:
  - NPV >> 0 = GO!!
  - NPV << 0 = No Go*
  - NPV = 0 ± ε = Think More*.

*Are there future options in project?

Takeaways

- It is important to try to **separate the value** created by ***underlying assets*** and those coming from ***financing*.**
- **APV** does this directly and **is a very flexible tool**. Sometimes it can be difficult because **you need to know the future levels of debt.**
- In APV it is clear the source of the value generation.
  - APV does not mix the valuation of operation with the effects of financing.
- **Both WACC and APV force you to get the asset beta, so you have to be able to do this!**

Takeaways (cont.)

- For APV we need to know the level of debt outstanding each year (very suitable for LBOs).
- For WACC we need to know the D/V ratio each year (very suitable if D/V is constant).
- You should understand the conceptual differences between these two methods and **why in principle we prefer APV**.
- Remember there are other cash flows methods you should bear in mind. Go through the examples and be sure you are capable of understanding each of the cash flows.
- **Best to use multiples as a check for the valuation based on discounted cash flows.**

Valuation and uncertainty

Uncertainty

- You might have the feeling the valuation is a kind of «black box»: you input numbers and you get a value.

- The value of any financial asset is the present value of **EXPECTED** future cash flows.
  - What if things don't go as expected?
  - It is much better to know upfront what is the result if things go differently, than wait till they happen and suffer the consequences.

- You should analyze possible deviations. Some variables and scenarios may have incredible impact on the value of our project or company.
  - These are the ones to test.
  - variables: discount rates, growth rates (especilaly perpetual growth rates), margins,...
  - Scenarios: pessimistic, optimistic,...

- **How do we deal with uncertainty?**

How do we deal with uncertainty?

In our valuation we should include analyses that show us how uncertainty can affect the value of our project or company. There are several tools for this purpose:

- **Sensitivity analysis:** changes one variable at a time and calculates its impact on the value.
- **Scenario analysis:** plans for differences in several variables at the same time (in a scenario) and calculates its impact in the value.
- **Monte-Carlo simulation:** is a computer-based technique that allows you to repeated random sampling in variables (more than 1,000) to model the probability of different outcomes in valuation.

Example

Example - assumptions

Example – Constant D/E

Assumptions:

- German company, with core business in Germany.
- Asset beta (unlevered beta) = 1 (based on an average of peer companies).
- Corporate tax rate = 25%
- D/E (constant) = 25%
- Initial revenues (31/12/2025) = 1,000
- Growth rate of revenues = 5% (2026); 4% (2027); 3.5% (2028); 3% (2029); 2.5% (2030) and 2% (2031).
- EBITDA margin = 40%
- PPE (gross value) in 2025 = 750
- D&A = 8.5% of gross PPE
- NWC = 20% of revenues
- Cost of debt = 3.5% (based on the market cost of long term debt of the company)
- Perpetual growth rate (nominal) = 2%

Example – operating cash flows (2026)

- Let's start by calculating the operating cash flow in 2026:
  - Revenues in 2026 = 1,050 (1,000 x 5%)
  - EBITDA in 2026 = 420 (1,050 x 40%)
  - D&A (*) = - 73 (855 x 8.5%)
  - EBIT = 347
  - Taxes = - 87 (347 x 25%)
  - **NOPAT = 260**

- (*) We need to calculate the depreciation: D&A = (gross PPE 2026) x 8.5% = (gross PPE 2025 + Capex2026) x 8.5% = (750 + 1050 x 10%) x 8.5% = (750 + 105) x 8.5% = 73

- **Operating Cash Flow** = NOPAT + D&A = 260 + 73 = **333**

Example – free cash flows (2026)

- **We need to calculate the investment cash flow in 2026: capital expenditures (CAPX) and change in net working capital (Δ NWC):**
  - Capex (2026)= 105 (1,050 x 10%)
  - Δ NWC (2026) = 10 (210 – 200)
- (*) Δ NWC = NWC(2026) – NWC(2025) = (1,050 x 20%) – (1,000 x 20%) = 210 – 200 = 10
- **Investment Cash Flow (2026)** = Capex (2026) + Δ NWC (2026) = 105 + 10 = **115**
- **Free Cash Flow (2026)** = Operating Cash Flow – Investment Cash Flow = 333 – 115 = **218**

Example – free cash flows

	2025	2026 F	2027 F	2028 F	2029 F	2030 F	2031 F	Perpetual
Revenues	1000	1 050	1 092	1 130	1 164	1 193	1 217	
Growth rate		*5,00%*	*4,00%*	*3,50%*	*3,00%*	*2,50%*	*2,00%*	
EBITDA	400	420	437	452	466	477	487	
EBITDA Margin	*40,00%*	*40,00%*	*40,00%*	*40,00%*	*40,00%*	*40,00%*	*40,00%*	
D&A	- 64	- 73	- 82	- 92	- 101	- 112	- 122	
% of gross PPE	*8,50%*	*8,50%*	*8,50%*	*8,50%*	*8,50%*	*8,50%*	*8,50%*	
PPE (gross value)	750	855	964	1 077	1 194	1 313	1 435	
EBIT	336	347	355	361	364	366	365	
Tax	- 84	- 87	- 89	- 90	- 91	- 91	- 91	
NOPAT	252	260	266	270	273	274	274	
(+) D&A	64	73	82	92	101	112	122	
Operating Cash Flow		**333**	**348**	**362**	**375**	**386**	**396**	
(-) Δ in NWC		- 10	- 8	- 8	- 7	- 6	- 5	
NWC	*200*	*210*	*218*	*226*	*233*	*239*	*243*	
% of sales	*20,00%*	*20,00%*	*20,00%*	*20,00%*	*20,00%*	*20,00%*	*20,00%*	
(-) Capex	- 100	- 105	- 109	- 113	- 116	- 119	- 122	
% of sales	*10,00%*	*10,00%*	*10,00%*	*10,00%*	*10,00%*	*10,00%*	*10,00%*	
Investment Cash Flow		**- 115**	**- 118**	**- 121**	**- 123**	**- 125**	**- 126**	
FCF		**218**	**230**	**241**	**251**	**261**	**269**	**275**

- **Perpetual Free Cash Flow =**

  Free Cash Flow (2031) x (1 + g) =

  269 x 1,02 = 275

Example – WACC

- **Discount rate (using CAPM) is the WACC: (D/V) x (1-τ) x r_D + (E/V) x r_E**
  - **D/V** = (D/E) / (1+ (D/E)) = 0.25 / (1 + 0.25) = **20%**
  - **τ = 25%**
  - **r_D = 3.5%**
    - $r_D = r_f + \beta_D$ x MRP ; 3.5% = 2.85% + β_D x 4.33% ; β_D = 0.15
  - **r_f = 2.85%** (yield of German bonds 10Y)
  - **MRP = 4.33%** (based on the information provided on the data base of professor Damodaran for Germany - Update July 2025)
  - **E/V** = 1- D/V = 1- 20% = **80%**
  - **r_E** = $r_f + \beta_E$ x MRP = 2.85% + 1.21 x 4.33% = **8.10%**
    - When D/V is constant: β_A = D/V x β_D + E/V x β_E ; 1.0 = 20% x 0.15 + 80% x β_E ; **β_E = 1.21**

- **WACC: (D/V) x (1-τ) x r_D + (E/V) x r_E = 20% x (1 - 25%) x 3.5% + 80% x 8.10% = 7.01%**

Example – EV using WACC

	2026 F	2027 F	2028 F	2029 F	2030 F	2031 F	Perpetual
FCF	218	230	241	251	261	269	275
Discount factor	1.07	1.15	1.23	1.31	1.40	1.50	

PV Free Cash Flows	1 159
PV Perpetuity	3 654
EV	**4 813**
Net Debt	**- 963**
Equity Value	**3 850**

- **Discount factor (2026)** = 1+ r_{WACC} = 1 + 7.01% = **1.07**
- **Discount factor (2027)** = $(1+ r_{WACC})^2$ = $(1 + 7.01\%)^2$ = **1.15**
- **Discount factor (2031)** = $(1+ r_{WACC})^6$ = $(1 + 7.01\%)^6$ = **1.50**
- **PV FCF (31/12/2025)** = FCF 2026/DF2026 + FCF2027/DF2027 +… +FCF2031/DF2031 = 218/1,07 + 230/1.15 + 241/1.23 + 251/1.31 +261/1.4 + 269/1.50 = **1,159**
- **PV Perpetuity (31/12/2025)** = (perpetual FCF / (r_{WACC} – g)) / DF2031 = (275 / (7.01% - 2%)) / 1.50 = 5,485 / 1.50 = **3,654**
- **Net Debt (31/12/2025)** = EV (31/12/2025) x D/V = 4,813 x 20% = **963**

Example – WACC

	2025	2026 F	2027 F	2028 F	2029 F	2030 F	2031 F	2032 F
EV (Present Value)	**4 813**	**4 932**	**5 047**	**5 159**	**5 269**	**5 377**	**5 485**	**5 595**
Net Debt	- 963	- 986	- 1 009	- 1 032	- 1 054	- 1 075	- 1 097	- 1 119
% of EV	*20,00%*	*20,00%*	*20,00%*	*20,00%*	*20,00%*	*20,00%*	*20,00%*	*20,00%*
Equity Value (Present Value)	**3 850**	**3 945**	**4 037**	**4 127**	**4 215**	**4 302**	**4 388**	**4 476**

EV, Equity Value and Net Debt in 30/12/2026:

- **New Discount factor (2027)** = (1+ r_{WACC}) = (1 + 7.01%) = **1.07**
- **New Discount factor (2031)** = $(1+ r_{WACC})^5$ = $(1 + 7.01\%)^6$ = **1.40**
- **PV FCF (31/12/2026)** = FCF2027/DC2027 +… +FCF2031/DF2031 = 230/1.07 + 241/1.23 + 251/1.31 +261/1.4 + 269/1.50 = **1,022**
- **PV Perpetuity (31/12/2026)** = (perpetual FCF / (r_{WACC} – g)) / 1.40 = (275 / (7.01% - 2%) / 1.50 = 5,485 / 1.40 = **3,910**
- **EV (31/12/2026)** = 1,022 + 3910 = **4,932**
- **Net Debt (31/12/2026)** = EV (31/12/2026) x D/V = = 4,932 x 20% = **986**

Example – APV

- **Recall: $V = V_U + PV(TS)$**

- **Free Cash Flow (2026)** = Operating Cash Flow – Investment Cash Flow = 333 – 115 = **218**

- $\mathbf{r_A} = r_f + \beta_A$ x MRP = 2.85% + 1.00 x 4.33% = **7.18%**

- **TS (2026)** = Interest x τ = Net Debt 2025 x r_D x τ = 963 x 3.5% x 25% = 34 x 0.25 = **8**

	2026 F	2027 F	2028 F	2029 F	2030 F	2031 F	Perpetual
FCF	**218**	**230**	**241**	**251**	**261**	**269**	**275**
Discount factor	1.07	1.15	1.23	1.32	1.41	1.52	
Interest	- 34	- 35	- 35	- 36	- 37	- 38	- 38
Tax Shields	**8**	**9**	**9**	**9**	**9**	**9**	**10**
Discount factor	1.07	1.15	1.23	1.32	1.41	1.52	

Example – APV

PV Free Cash Flows (all equity)	1 153
PV Perpetuity (all equity)	3 496
PV (all equity)	**4 648**
PV tax shields cash flows	42
PV Perpetuity	122
PV Tax Shields	**164**
EV	**4 813**
Net Debt	**- 963**
Equity Value	**3 850**

- **Discount factor (2026)** = 1+ r_A = 1 + 7.18% = **1.07**
- **Discount factor (2027)** = $(1+ r_A)^2$ = $(1 + 7.18\%)^2$ = **1.15**
- **Discount factor (2031)** = $(1+ r_A)^6$ = $(1 + 7.18\%)^6$ = **1.52**
- **PV FCF (31/12/2025)** = FCF2026/DF2026 + FCF2027/DF2027 +… +FCF2031/DF2031 = 218/1,07 + 230/1.15 + 241/1.23 + 251/1.32 +261/1.41 + 269/1.52 = **1,153**
- **PV FCF Perpetuity (31/12/2025)** = (perpetual FCF / (r_{WACC} – g)) / DF2031 = (275 / (7.18% - 2%) / 1.50 = 5,300 / 1.52 = **3,496**
- **PV TS (31/12/2025)** = TS 2026/DF2026 + FCF2027/DF2027 +… +FCF2031/DF2031 = 8/1,07 + 9/1.15 + 9/1.23 + 9/1.32 + 9/1.41 + 9/1.52 = **42**
- **PV TS Perpetuity (31/12/2025)** = (perpetual TS / (r_A – g)) / DF2031 = (10 / (7.18% - 2%)) / 1.52 = 5,300 / 1.50 = **164**
- **EV (31/12/2026)** = VU + PV(TS) = 4,648 + 164 = **4,813**
- **Net Debt (31/12/2025)** = EV (31/12/2025) x D/V = 4,813 x 20% = **963**

Example – CCF

- **Recall: CCF = FCF + Interest tax shield**
- **Capital Cash Flow (2026)** = FCF + Interest tax shield = 218 + 8 = **226**
- $\mathbf{r_A} = r_f + \beta_A$ x MRP = 2.85% + 1.00 x 4.33% = **7.18%**

Example – CCF

	2026 F	2027 F	2028 F	2029 F	2030 F	2031 F	Perpetual
FCF	**218**	**230**	**241**	**251**	**261**	**269**	**275**
Interest tax shields	8	9	9	9	9	9	10
Capital Cash Flow	**227**	**239**	**250**	**260**	**270**	**279**	**284**
Discount factor	1.07	1.15	1.23	1.32	1.41	1.52	

PV Capital Cash Flows	1 195
PV Perpetuity	3 618
EV	**4 813**
Net Debt	**- 963**
Equity Value	**3 850**

- **Discount factor (2026)** = 1+ r_A = 1 + 7.18% = **1.07**
- **Discount factor (2031)** = $(1+ r_A)^6$ = $(1 + 7.18\%)^6$ = **1.52**
- **PV CCF (31/12/2025)** = CCF2026/DF2026 + CCF2027/DF2027 +… +CCF2031/DF2031 = 227/1,07 + 239/1.15 + 250/1.23 + 260/1.32 +270/1.41 + 279/1.52 = **1,195**
- **PV CCF Perpetuity (31/12/2025)** = (perpetual CCF / (r_A – g)) / DF2031

  = (284 / (7.18% - 2%)) / 1.50 = 5,485 / 1.52 = **3,618**
- **Net Debt (31/12/2025)** = EV (31/12/2025) x D/V = 4,813 x 20% = **963**

Example – ECF

- **Recall: ECF = CCF – Interest + Change in Debt = FCF + Interest tax shield - Interest + Change in Debt**
- **Change in Debt (2026)** = Net Debt 2026 – Net Debt 2025 = 986 – 963 = **24**
- **Capital Cash Flow (2026)** = FCF + Interest tax shield – Interest + Change in Debt= 218 + 8 – 34 + 24 = **216**
- $\mathbf{r_E} = r_f + \beta_E$ x MRP = 2.85% + 1.21 x 4.33% = **8.10%**

Example – CCF

	2026 F	2027 F	2028 F	2029 F	2030 F	2031 F	Perpetual
FCF	**218**	**230**	**241**	**251**	**261**	**269**	**275**
Interest tax shields	8	9	9	9	9	9	10
Interest	- 34	- 35	- 35	- 36	- 37	- 38	-39
Change in Debt	24	23	22	22	22	22	22
Equity Cash Flow	**217**	**228**	**237**	**246**	**255**	**262**	**268**
Discount factor	1.08	1.17	1.26	1.37	1.48	1.60	

PV Equity Cash Flows	1 100
PV Perpetuity	2 750
Equity Value	**3 850**
Net Debt	**963**
EV	**4 813**

- **Discount factor (2026)** = 1+ r_E = 1 + 8.10% = **1.08**
- **Discount factor (2031)** = $(1+ r_E)^6$ = $(1 + 8.10\%)^6$ = **1.60**
- **PV ECF (31/12/2025)** = ECF2026/DF2026 + ECF2027/DF2027 +… +ECF2031/DF2031 = 217/1,08 + 228/1.17 + 237/1.26 + 246/1.37 +255/1.48 + 262/1.60 = **1,100**
- **PV ECF Perpetuity (31/12/2025)** = (perpetual ECF / (r_E – g)) / DF2031

  = (268 / (8.10% - 2%)) / 1.60 = 4,388 / 1.60 = **2,750**

Appendix:
Computing Asset Betas

Computing Asset Betas

How to Measure β_A?

- The value of the levered firm is given by $V_L = D + E$.
- Since, $V_L = V_U + PV(TS)$, then $V_U = D + E - PV(TS)$.
- And therefore:

$$V_U \beta_A = D\beta_D + E\beta_E - PV(TS)\beta_{TS}$$

- Which we can be re-written as:

$$\beta_A = \frac{D}{V_U}\beta_D + \frac{E}{V_U}\beta_E - \frac{PV(TS)}{V_U}\beta_{TS}$$

- Now we need to assume something about β_{TS}.

Computing Asset Betas (cont.)

$$\beta_A = \frac{D}{V_U}\beta_D + \frac{E}{V_U}\beta_E - \frac{PV(TS)}{V_U}\beta_{TS}$$

- **Assumption 1:** The risk of the tax savings is the same as the risk of the debt that generates it ($\beta_{TS} = \beta_D$).
- Then the previous equation becomes:

$$\beta_A = \frac{D}{V_U}\beta_D + \frac{E}{V_U}\beta_E - \frac{PV(TS)}{V_U}\beta_D$$

- And since $V_U = V - PV(TS)$, we have:

$$\beta_A = \frac{D - PV(TS)}{V - PV(TS)}\beta_D + \frac{E}{V - PV(TS)}\beta_E$$

Computing Asset Betas (cont.)

- This formula simplifies under some special cases.
  - If D is constant:

$$\beta_A = \frac{D\,(1-\tau_C)}{V-\tau_C D}\beta_D + \frac{E}{V-\tau_C D}\beta_E$$

  - If, in addition, the debt is riskless (b_D = 0):

$$\beta_A = \frac{\beta_E}{\left(1+(1-\tau_C)\frac{D}{E}\right)}$$

Computing Asset Betas (cont.)

$$V_U\beta_A = D\beta_D + E\beta_E - PV(TS)\beta_{TS}$$

- **Assumption 2:** The risk of the tax savings is the same as the risk of the existing assets ($\beta_{TS} = \beta_A$).
- Then we are left with,

$$\big(V_U + PV(TS)\big)\beta_A = D\beta_D + E\beta_E$$

- And since, $V_L = V_U + PV(TS)$ we have that:

$$\beta_A = \frac{D}{V}\beta_D + \frac{E}{V}\beta_E$$